



Effectiveness Validation of Visualization Designs

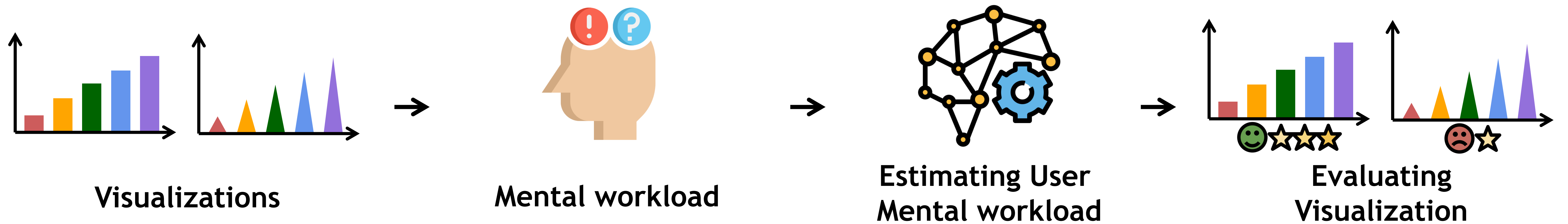
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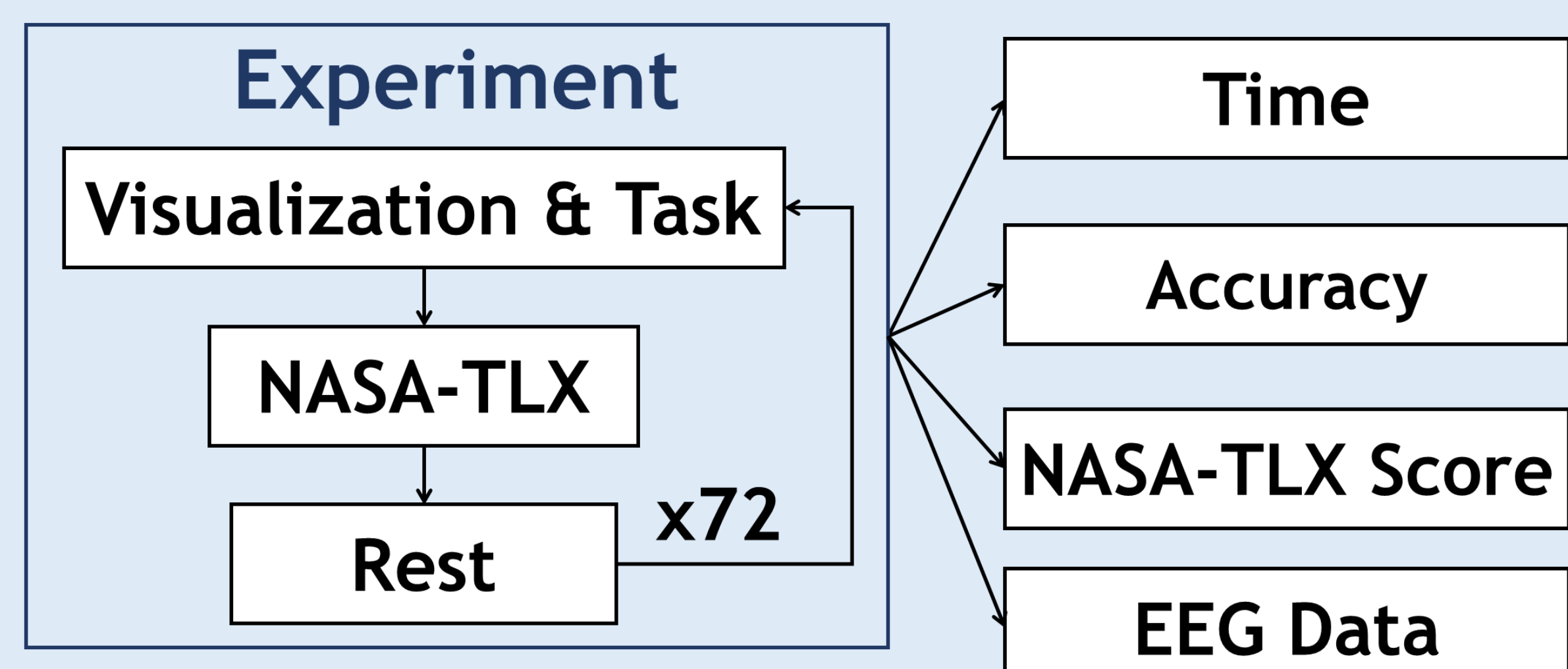
Motivation

- We examine whether there is a **difference** in **mental workload** for the **visualization designs**
- We propose a **mental workload estimation model** specialized for each individual to evaluate visualization designs using EEG.

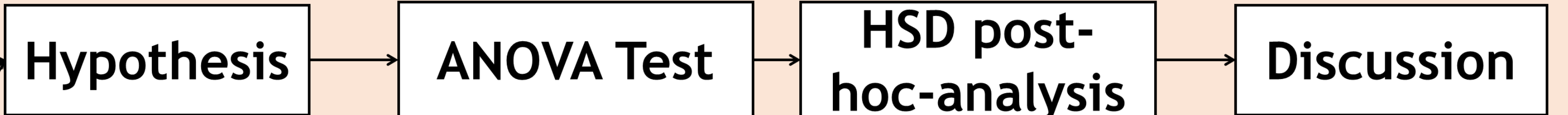
System Process



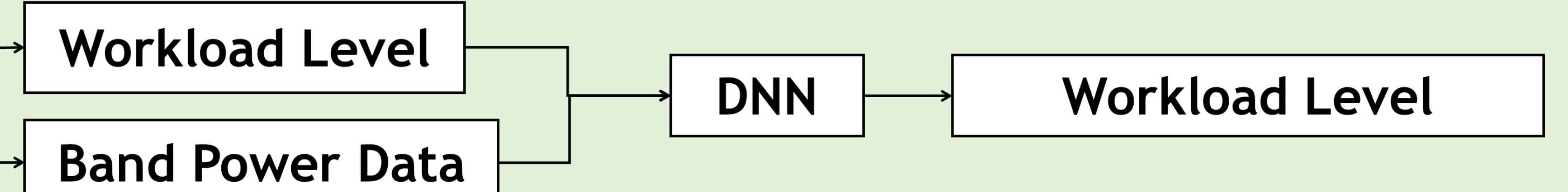
Data Collection



Design Guide Influence



Mental Workload Estimation Model



Our work procedure

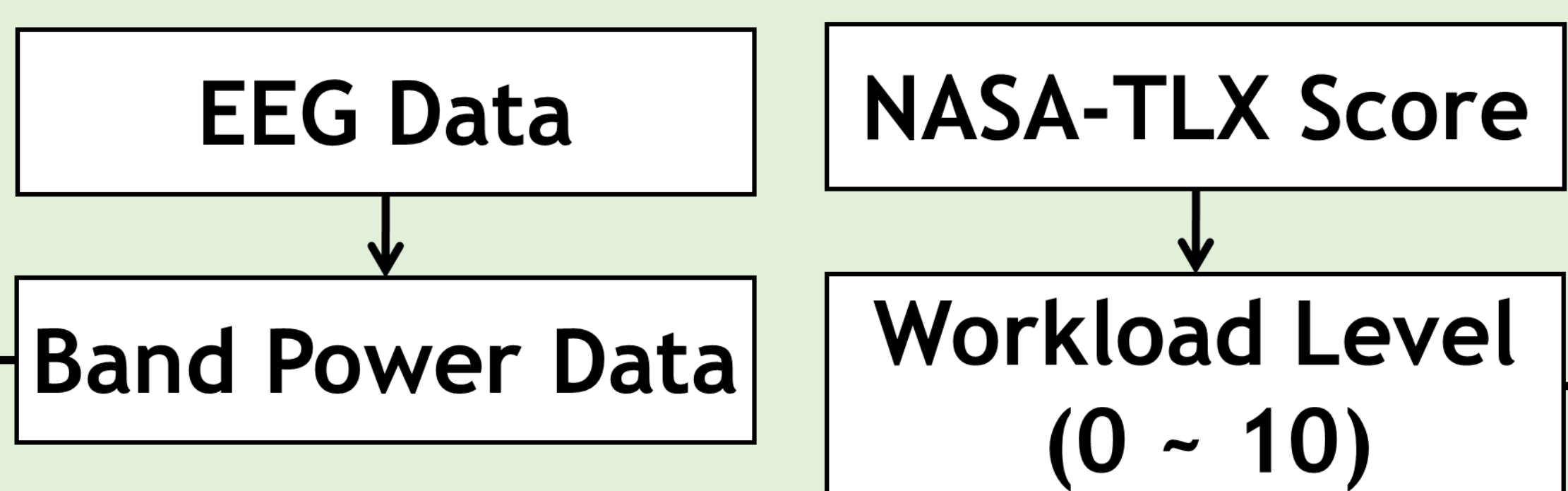
Data Collection

- The participant uses the visualization to answer the visualization task and records the answer.
- The participant takes a NASA-TLX survey and takes a break.
- While the experiment, the response time, accuracy, NASA-TLX score, and EEG data are collected.

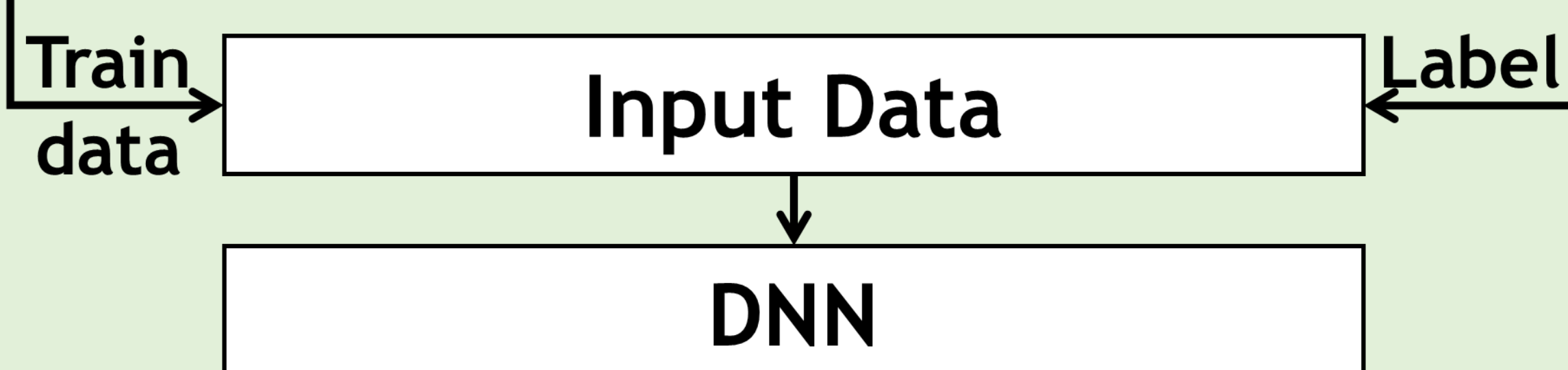
Design Guide Influence

- According to the ANOVA test, **accuracy** and **NASA-TLX** are **affected** by the **visualization designs**, but time is not affected.
- According to the Turkey HSD, the designs for **Spatial Position & Shape** and **Color** design have **different results** in accuracy and NASA-TLX **compared to the Base**.

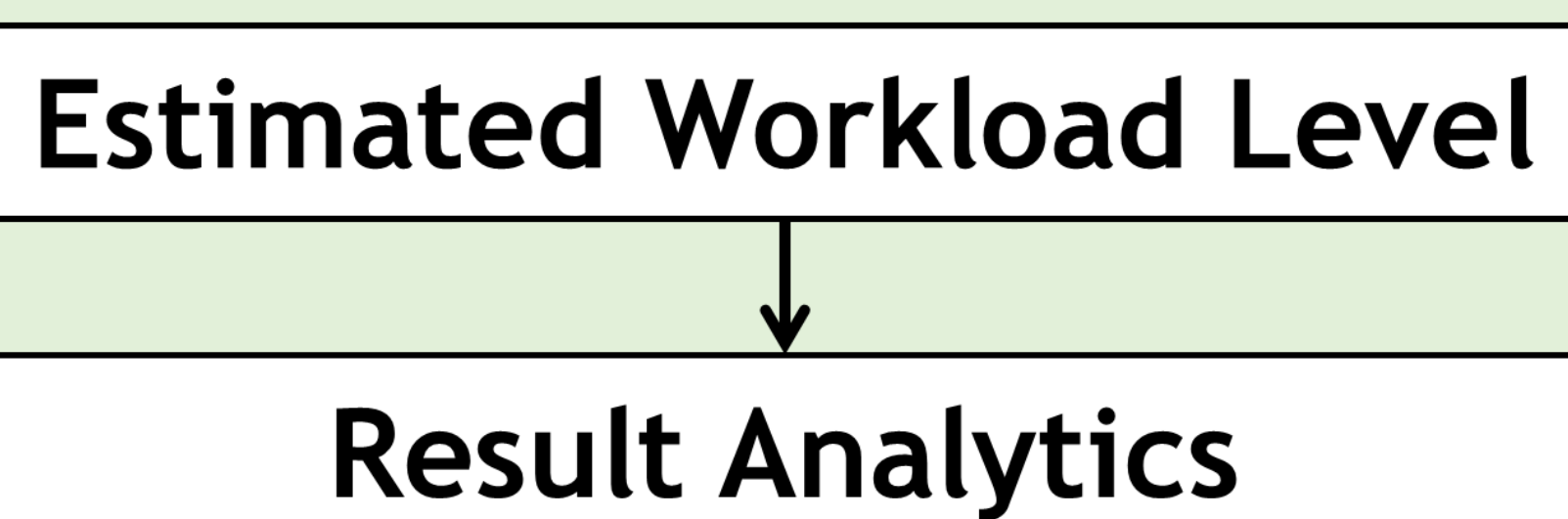
Preprocessing



Model Learning



Workload Estimation



Mental workload estimation model

Mental Workload Estimation Model

- We present our mental workload estimation model using EEG data specialized for each individual to evaluate visualization designs.
- We estimate the mental workload level utilizing Deep Neural Network (**DNN**).
- The proposed model utilize the mental workload level as a label and band power data as train data.
- Our proposed DNN model predicts the mental workload levels with errors between 0~0.2.

Conclusion

- In this paper, we utilized mental workload, response time, and accuracy to check whether visualization design guidelines were meaningful.
- Also, we proposed a mental workload estimation model specialized for each individual to evaluate visualization design using EEG data.